

Question 6 – Electrons in Electric and Magnetic Fields

(a)(i) Draw the electric field pattern between cathode and anode

✓ 1 mark

✓ **Answer:**

- Field lines go **left to right**, from positive (anode) to negative (cathode)
- Lines should be **straight, equally spaced**, and **perpendicular** to both surfaces
- Minimum of **three lines**: top, middle, and bottom

 *Tip:* Electric field lines always go from **+** to **–**, and show the **direction a positive charge would move**.

(a)(ii) Calculate the maximum speed of an electron at the anode

✓ 3 marks

Given:

- Gain in kinetic energy $E_k = 3.84 \times 10^{-16} \text{ J}$
- Mass of electron $m = 9.11 \times 10^{-31} \text{ kg}$

$$E_k = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 3.84 \times 10^{-16}}{9.11 \times 10^{-31}}} = 2.90 \times 10^7 \text{ m/s}$$

(a)(iii) Calculate the maximum acceleration between cathode and anode

✓ 3 marks

Distance between cathode and anode: $s = 45.0 \text{ mm} = 0.0450 \text{ m}$

Use:

$$v^2 = u^2 + 2as \Rightarrow a = \frac{v^2}{2s} = \frac{(2.90 \times 10^7)^2}{2 \times 0.0450} = 9.34 \times 10^{15} \text{ m/s}^2$$

(b) Direction of the magnetic field

✓ 1 mark

From the right-hand rule:

- Electrons are **negatively charged**, so reverse the force direction
- If electrons bend **downward**, and motion is to the **right**, then the **magnetic field must be OUT of the page**.

✓ **Answer:**

The direction of the magnetic field is **out of the page**.

 Revision Tips

- Electric field = straight lines from + to –
- Electrons **accelerate** due to electric potential (E field), then **curve** in magnetic fields
- Use $E_k = \frac{1}{2}mv^2$ and $v^2 = u^2 + 2as$ where applicable
- **Right-hand rule** applies to **positive charges** — reverse for electrons!